# An Explainable AI-Based Decision Support System for Dynamic Water Quality Index Prediction and Recommendation

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**Abstract**

Water quality is characterized by multiple physical and chemical parameters. This study presents an explainable AI-based decision-support system for estimating the constructed station-year Water Quality Index (WQI) from six predictors: BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH. Five parameters, excluding DO, construct the target. The study uses Northern Ireland Environment Agency (NIEA) observations from 1990 to 2024; it estimates the index for the represented station-year rather than forecasting future water quality.

The work includes data preprocessing, missing-value analysis, WQI construction, machine learning and deep learning model comparison, model validation, explainable AI and recommendation generation. Random Forest, Extra Trees, Gradient Boosting, XG-Boost, and Cat-Boost are five ML-models were compared with Deep Neural Networks and Tab-Net. This models are evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and $R^2$.

By using of Cat-Boost model showed strong performance and also selected for the explainability and decision support stages. Fivefold cross-validation are performed to check the consistency of the Cat-Boost model. To understand how the input parameters are contributed to individual predictions and to Study the overall importance of the features SHAP model is used. By combining the predicted WQI, parameter values, model explanations and guideline exceedance A recommendation engine was developed.

The complete system was implemented using Stream-Lit. In dashboard section the user should enter the water quality parameters as input and receive the predicted WQI, water-quality category, guideline analysis, SHAP explanation

and recommendation. Therefore, The system connects water-quality prediction with explainability, practical decision support in a single workflow.

**Keywords:** Water Quality Index, Water Quality Assessment, Water Quality Prediction, Machine Learning, Deep Learning, Explainable Artificial Intelligence, SHAP, Decision Support System

# 1 Introduction

Water is one of the most important natural resources for human life, ecological systems, agriculture, and industrial activities also. The quality of water directly influences the following aquatic ecosystems, public health, and the sustainable management of water resources. However, water quality can also change substantially because of the some natural processes and human activities. Agricultural runoff, wastewater discharge, industrial activities, urban development, and the changes in the environmental conditions can also cause different pollutants into rivers and other water bodies. At the same time, natural processes such as the seasonal variation, hydrological changes, and then the geological characteristics can also cause the measured properties of water. For these reasons, continuous and systematic water-quality monitoring is very important for the understanding of changes in water conditions and having a good effective environmental management. Water quality is generally analyzed using a combination of physical, chemical, and biological parameters. Individual parameters provide some information about the different characteristics of a water body. Biochemical oxygen demand (BOD) is associated with the amount of oxygen that is required for the following biological decomposition of organic matter, while the dissolved oxygen (DO) provides the some information about the oxygen conditions that are available in the aquatic environment. Nitrogen-related parameters such as the ammonia-nitrogen, nitrite-nitrogen, and nitrate-nitrogen can also provide some information about the nutrient loading and the possible pollution in it. Similarly, pH represents the acidity or alkalinity of the water and can also affect the chemical reactions and the aquatic organisms. Although each parameter provides some useful information, interpreting a large number of measurements independently can be kind of difficult, particularly when the observations are collected from many other monitoring locations over the long periods. A large monitoring dataset contain thousands of observations with spatial and temporal variation, making it difficult to obtain a clear representation of the overall water-quality condition from individual measurements we have.Water Quality Index (WQI) methods have been developed to address this problem by combining some information from multiple water-quality parameters into a single numerical value index. A WQI provides a clear representation of overall water quality and can make the water-quality information easier for us to communicate and interpret. Different WQI approaches have been proposed using some different parameter sets, reference objectives, weighting methods, normalization procedures, and aggregation strategies also. Consequently, the same monitoring data can produce different index values depending on selected WQI formulation. It is important to clearly define how the index is constructed and

what the resulting value actually represents. In our study, the Canadian Council of Ministers of the Environment Water Quality Index (CCME-WQI) framework is used to construct the following station-year water-quality target. The CCME-WQI considers the scope, frequency, and amplitude of exceedances relative to some defined objectives, providing a representation of water-quality conditions. The WQI used in study is a constructed index rather than a directly measured variable, and the prediction task is designed to estimate this constructed station-year index. The availability of the long-term environmental monitoring datasets has created some opportunities to apply data-driven methods to water-quality assessment. Conventional assessment approaches generally require some repeated examination of multiple parameters and become increasingly difficult as the volume and dimensionality of monitoring data increase further. Machine learning (ML) methods provides some alternative approach by learning the relationships between the input variables and a target variable from observed data. In water-quality applications, regression algorithms can be used to identify nonlinear relationships among chemical and physical parameters and overall water-quality indicator. Previous studies have investigated some different machine-learning techniques for water-quality prediction and WQI estimation, demonstrating that ensemble learning methods can be useful for modelling complex relationships between the following water-quality variables. However, model performance is dependent on some several factors, including the characteristics of the dataset, preprocessing procedures, selected predictors, target definition, and the experimental design. Differences in these kind of factors can make direct comparison between studies kind of difficult. Another important challenge in the environmental machine learning is that the interpretation of model predictions. A model that produce a highly accurate numerical prediction, but prediction accuracy alone does not explain why a particular result was obtained at that time. This is important in water-quality assessment because users need to understand which parameters are contributing to poor or good predicted condition.For example, a model predicts a low WQI,then the numerical prediction itself does not tell that the prediction is primarily associated with the nitrate, nitrite, ammonia, BOD, pH, or any other input variables. Without an interpretable mechanism, a machine-learning model can remain difficult for us to use in practical environmental decision-making. Explainable Artificial Intelligence (XAI) methods can help check this issue by providing information about the contribution of the individual input features to model predictions. In our study, SHapley Additive exPlanations (SHAP) are used to interpret the following selected prediction model and examine how the individual water-quality parameters contribute to the predicted WQI values. The need for interpretation becomes even more important when machine-learning models are intended to support the practical decisions rather than only producing some numerical predictions. A predicted WQI can indicate the estimated quality category of station-year condition, while SHAP can provide some information about the parameters contributing to the prediction. However, prediction and explanation alone still not provide sufficient information for a user who needs to determine what should be examined next further. A decision-support system should connect the numerical prediction with parameter-level assessment and an understandable interpretation of the observed

condition. This motivates an integrated workflow process in which water-quality measurements are first processed to obtain the model inputs, the WQI is predicted, the prediction is explained using SHAP, potentially important parameters are assessed against the selected guideline objectives, and the further structured recommendations are generated. The workflow can be represented as: Water-quality measurements → WQI prediction → SHAP explanation → parameter assessment → recommendation Such an integrated approach wants to bridge the gap between predictive modelling and practical interpretation rather than treating prediction, explainability, and decision support as some separate tasks. Previous research has demonstrated the value of machine learning for water-quality prediction and the usefulness of XAI for interpreting model behaviour, but differences in the following datasets, preprocessing strategies, target definitions, evaluation procedures, and validation designs remain some important considerations. Even some high predictive accuracy does not by itself establish geographic or temporal generalization. These considerations are particularly relevant when the models are further developed using environmental monitoring data collected from multiple locations and year The study addresses these following challenges through an integrated explainable artificial intelligence framework using the long-term river-water-quality observations from the Northern Ireland Environment Agency (NIEA). The project dataset covers observations from 1990 to 2024 and includes measurements from a large number of monitoring stations. The long observation period and the multiple monitoring locations provide a heterogeneous dataset containing the variation across both the stations and years. From the available water-quality measurements, BOD, DO, ammonia-nitrogen, nitrite-nitrogen, nitrate-nitrogen, and pH are used as predictors for the machine-learning and deep-learning models. Five of these parameters—BOD, ammonia-nitrogen, nitrite-nitrogen, nitrate-nitrogen, and pH—are used to construct the CCME-WQI target, while the DO is included as a predictor in the modelling stage. The prediction task estimates the needed constructed station-year WQI from six predictor variables rather than some forecasting a future water-quality condition. An important component of the following proposed workflow is the treatment of the incomplete monitoring observations. Long-term environmental datasets commonly contain missing measurements because monitoring frequency can vary between some stations and parameters, and individual measurements can not be available for every sampling event. Ignoring missingness without assessment can reduce the number of usable observations and potentially affect following WQI construction and predictive modelling. So, the present study includes a missing-value analysis and applies the parameter-specific imputation procedures to recover the selected missing measurements. The resulting datasets are evaluated through separate baseline and imputed modelling branches, allowing the effect of the preprocessing strategy to be examined further rather than assuming that some single preprocessing choice is universally optimal. This provides more transparent basis for evaluating the following WQI construction and the predictive models. Following preprocessing and WQI construction, several machine-learning and deep-learning approaches are then compared under a common experimental framework. The machine-learning models include Random Forest, Extra Trees, Gradient Boosting, XGBoost, and CatBoost, while Deep Neural Networks (DNN) and TabNet are included as the following deep-learning approaches.

The models are then evaluated using the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination ($R^2$). Using the same target formulation, predictor set, train-test split, and then the evaluation metrics provides us a consistent basis for comparing different modelling approaches. The comparison is to identify a suitable predictive model for the further explainability and decision-support stages rather than assuming that a algorithm will always perform best across the environmental datasets. Among the evaluated models, CatBoost is selected for the explainability and decision-support stages based on its strong predictive performance in the experimental comparison we did. Its performance is further then examined using five-fold cross-validation to find the consistency of the model across different subsets of the available station-year records. SHAP analysis is then applied to the selected CatBoost model to find the contribution of the six predictor variables to individual WQI predictions. This provides us both a global perspective on the feature importance and a local perspective on how individual measurements influence some specific predictions.These model-derived contributions are interpreted as the predictive associations rather than some direct evidence of environmental causation. The distinction is important because an explanatory model can indicate which variables influence the model output without proving that a particular variable is the physical cause of the following observed water-quality condition. The final component of the our proposed framework is recommendation-oriented decision-support layer. The system combines the predicted WQI category, input parameter values, selected guideline exceedance information, and model explanations to generate some structured recommendations for it. The purpose of these recommendations is not to replace environmental investigation or regulatory decision-making, but to provide some interpretable level assessment that can help users find parameters requiring attention and consider appropriate monitoring or some source-investigation actions. This distinction is important because a machine-learning prediction and a guideline exceedance do not by themselves establish the exact source of some pollution or prescribe a scientifically validated remediation action as needed. Finally, the recommendation component is designed as the decision support rather than as some autonomous environmental management system. The complete workflow is then implemented through an interactive Streamlit dashboard. The dashboard allows a user to enter the six water-quality predictor values and obtain a predicted WQI, corresponding water-quality category, parameter-level guideline assessment, SHAP-based explanation, and structured recommendation as output. By integrating these components into a single interface, the system moves beyond a some conventional prediction-only application. Instead, it provides a sequence from measurement input to prediction, explanation, assessment, and then the recommendation. This design is used to make the output of the machine-learning model easier to understand for users who not have direct access to the underlying model implementation. Overall, Our study develops an explainable AI-based decision-support framework for station-year WQI estimation using the following long-term NIEA river-water-quality observations. The work combines the data preprocessing and missing-value analysis, CCME-WQI construction, comparative machine-learning and deep-learning modelling, CatBoost validation, SHAP-based interpretation, parameter-level assessment, recommendation generation, and then interactive dashboard deployment . The

central contribution is not limited to obtaining a numerical WQI prediction. Instead, the study says to connect water-quality data, index estimation, predictive modelling, model interpretation, and practical decision support within a single overall workflow. By maintaining explicit distinction between the constructed WQI estimation, model-derived explanation, guideline assessment, and recommendation, the following proposed framework provides a transparent basis for us to use machine learning to water-quality assessment and the monitoring.

# 2 Related Work / Literature Review

## 2.1 Traditional Water Quality Index Approaches

Over the years, there are many WQI methodologies were developed, each addresses different parameter sets, reference standards, normalization techniques, weighting strategies and aggregation procedures. Consequently, identical water-quality dataset may results in different index values based on the WQI formulation that we selected.

The WQI is a constructed composite indicator rather than a directly measured variable. Yan et al. [7] discuss uncertainties in WQI models and limitations of machine-learning applications. In this study, the chosen formulation and objectives define the constructed target that the regression models estimate.

The Canadian Council of Ministers of the Environment Water Quality Index (CCME-WQI) incorporates scope, frequency, and amplitude of objective exceedances [12]. This study applies the framework to Northern Ireland Environment Agency (NIEA) observations [11], using the parameter set and objectives configured in the project implementation [13]; their status as statutory limits has not been established.

## 2.2 Machine Learning for Water Quality Prediction

Machine-learning methods have been studied for WQI estimation. Choudhary et al. [8] combined six regression learners in a stacked ensemble and used SHAP to interpret model predictions. This verified example motivates comparing regression models while keeping the target formulation and explanation scope explicit.

Yan et al. [7] review machine-learning approaches for water-quality prediction, including indicator and index prediction, and discuss limitations concerning data processing, model uncertainty, and generalization. This review does not establish geographic transfer of the models developed in the present project.

There are few limitations:

- Model performance is highly dependent on the characteristics of the dataset
- Predictive accuracy may vary across different geographic locations
- Spatial and Temporal generalization remains a critical concern
- High numerical performance does not necessarily equate to scientific validity
- Interpretability of the models is also essential.

## 2.3 Deep Learning for Water Quality Prediction

Deep Learning models have also gained attention in the prediction of water-quality because the water-quality conditions are been influenced by various environmental factors that interacts in complex ways.

Li et al. [6], in a 2024 review published in Environmental Science and Pollution Research, examine deep-learning methods for water-quality prediction and forecasting, including single-model and mixed-model approaches, optimization, and data fusion. Their review discusses data scarcity and long-term prediction challenges; these broader forecasting applications are distinct from the station-year index estimation implemented here.

Performance depends on: Size of dataset, Feature characteristics, Temporal and Spatial structures, Model architecture, preprocessing procedures, Hyperparameter configuration.

ML vs DL should be experimentally compared under the same evaluation conditions.

## 2.4 Explainable Artificial Intelligence and SHAP

Machine Learning models provides accurate WQI predictions, but their outputs or outcomes are often difficult to interpret. In water-quality applications, understanding the factors that influence model predictions is important, as these results can support environmental assessment and water-resource management.

Explainable AI (XAI) methods help describe model predictions. SHAP provides feature attributions to individual predictions [1], and Choudhary et al. [8] apply it to WQI regression. Pandey et al. [9] identify explainable AI as a direction for further water-quality prediction research. These attributions explain model outputs, not physical causality.

## 2.5 Decision-Support Systems

Previous studies have shown that ML and DL models can be used to predict water quality, while XAI methods can provide insight into the parameters influencing these predictions. However, prediction and explanation alone do not form a complete decision-support framework. This provides the motivation for developing an integrated approach that combines prediction, explainability, and decision-oriented analysis.

The aim is to connect constructed station-year WQI estimates and model explanations within a decision-support workflow. Malisaba et al. [10] discuss methodological and operational gaps, including reliance on localized datasets and limited assessment beyond accuracy metrics. These concerns motivate transparent reporting of the validation and advisory limitations of the present system.

# 3 Materials and Methods

***Proposed Methodology***

Figure 1 presents the project workflow linking constructed station-year WQI estimation, model explanation, and advisory outputs. The six-predictor regression task is index estimation, not future forecasting.




**Fig. 1** System architecture of the proposed explainable AI-based water quality decision support system.

## 3.1 Dataset Description

Our study uses long-term river-water-quality monitoring data from the Northern Ireland Environment Agency (NIEA). The public source is catalogued as *River Water Quality Monitoring 1990 to 2024 – All Parameters* [11]. The project dataset covers 1990 to 2024 and contains observations from multiple river monitoring stations across Northern Ireland. The original project dataset has 178,680 observations and 40 variables, including station information, geographic information, sampling dates, monitoring conditions, water-quality measurements, and associated quality indicators.

The dataset contains observations from the 1,311 monitoring stations, having both spatial and temporal variation in water-quality conditions.

## 3.2 Data Preprocessing

The data types, missing values, parameter ranges, station identifiers, and temporal coverage were examined before performing further preprocessing.

The sampling dates ranges from January 1990 to December 2024, while the monitoring records are distributed across some 1,311 stations.

### 3.2.1 Data Cleaning and Transformation

The baseline preprocessing script converts Date and extracts Year. It retains 13 columns: StationCode, Location, PrimaryBasin, GridReference, Date, Time, Year, and the six measurements BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH. Exactly 3,065 rows have all six measurements missing, leaving 175,615 rows in the saved core dataset. Rows with only some measurements missing are retained. The script does not delete rows because of extreme values, quality flags, duplicates, or individual missing parameters.

The final imputation script instead reads the original 178,680 rows, derives Year and Month, and retains all rows in its saved output. It does not apply the baseline all-six-missing filter. The imputation flags identify 10,178 recovered parameter measurements; this is a count of filled measurement cells, not additional observation rows.

Missing values were have not blindly replaced with the zero, since zero represent a valid measurement for some parameters and can introduce artificial observations.

### 3.2.2 Quality Control

The extreme-value script reports measurements above its inspection thresholds but does not remove them. The value-validation script reports negative-value and duplicate counts without dropping rows; the supplied core dataset has no negative values in the six selected measurements and no duplicate rows.

The original NIEA quality-flag fields are inspected by reporting their frequencies among observed measurements. These include $<$ and $>$ qualifiers. No flag-based exclusion, qualifier-dependent adjustment, or censoring correction is implemented in the supplied WQI-construction scripts.

## 3.3 Missing-Value Analysis and Imputation

Missing measurements are omitted separately for each parameter, rather than applying complete-case deletion across the five WQI parameters.

DO is not imputed by this script.

For each regression target, training rows must contain an observed target and nonmissing values for all its predictors. The regressor is instantiated with `n_estimators=100`, `random_state=42`, and `n_jobs=-1`; no other hyperparameters are explicitly set. Predictions replace only missing target values for rows with complete predictor values. The targets are processed sequentially in the order BOD, Ammonia-N, and Nitrite-N, so a later imputation stage can use values recovered by an earlier stage. Observed values are not overwritten.

For Nitrate-N and pH, observations are sorted by StationCode and Date. Within each station, the immediately preceding and immediately following values are obtained. A missing value is replaced with the arithmetic mean of these two values only when both are available. This is an adjacent-record average, not time-distance-weighted interpolation. Missing values without two available adjacent observations remain missing; no extrapolation or iterative filling is implemented.

The two approaches are evaluated independently using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and $R^2$.

## 3.4 CCME-WQI Construction

The Canadian Council of Ministers of the Environment Water Quality Index (CCME-WQI) framework combines multiple parameters and repeated observations through scope (F1), frequency (F2), and amplitude (F3) [12].

### 3.4.1 Parameter Selection

DO is included among the six ML predictors but excluded from the five-parameter WQI target.

**Table 1** Selected parameter-specific imputation approaches

| **Parameter** | Selected approach |
|---|---|
| BOD | Random Forest regression using Ammonia-N, Nitrite-N, Nitrate-N, pH, Year, and Month. |
| Ammonia-N | Random Forest regression using BOD, Nitrite-N, Nitrate-N, pH, Year, and Month. |
| Nitrite-N | Random Forest regression using BOD, Ammonia-N, Nitrate-N, pH, Year, and Month. |
| Nitrate-N | Arithmetic mean of the immediately preceding and following nonmissing adjacent observations in the same station. |
| pH | Arithmetic mean of the immediately preceding and following nonmissing adjacent observations in the same station. |

These parameters were selected based on availability in the NIEA dataset and their relevance to the assessment of organic pollution, nutrient contamination, and general chemical condition.

Table 3 lists the study-defined water-quality objectives used for CCME-WQI construction and guideline-exceedance analysis, together with the minimum-data criteria documented in the uploaded WQI construction scripts [13]. These are implemented study objectives, not verified statutory limits or a claim that CCME prescribes these particular values.

**Table 2** Parameters selected for WQI construction

| Parameter | Description |
|---|---|
| BOD | Biochemical Oxygen Demand |
| Ammonia-N | Ammonia Nitrogen |
| Nitrite-N | Nitrite Nitrogen |
| Nitrate-N | Nitrate Nitrogen |
| pH | Acidity or alkalinity indicator |

**Table 3** WQI thresholds and minimum-data criteria

| Parameter / Criterion | Threshold / Requirement |
|---|---|
| BOD | 3.0 mg/L |
| Ammonia-N | 0.05 mg/L |
| Nitrite-N | 0.5 mg/L |
| Nitrate-N | 0.5 mg/L |
| pH | 6.2–7.3 |
| Minimum Station-Year observations | 10 |
| Minimum available WQI parameters | 4/5 |

### 3.4.2 Station-Year Aggregation

Both WQI builders group individual observations by StationCode and Year. The supplied inputs contain no missing StationCode or Year values, so no rows are lost through missing grouping keys. The baseline group-size median is 12 and the mean is approximately 11.33.

The minimum observation requirement is implemented as `len(group) >= 10`. It counts rows in the input group, not complete measurement vectors or valid tests for an individual parameter. Consequently, a retained row with all WQI measurements missing still contributes to the imputed-input observation count, although it contributes no parameter tests. A group must also contain at least four available parameters out of BOD, Ammonia-N, Nitrite-N, Nitrate-N, and pH. Availability requires at least one nonmissing value for that parameter within the group. DO does not contribute to WQI parameter availability.

The five WQI parameters are not reduced to medians or means before computing the index. For each eligible group, the builder iterates over all nonmissing measurements of each parameter. Each available measurement contributes one test, and each guideline failure contributes a failed test and an excursion. A parameter is counted as failed if any of its available measurements fails its objective. Across the eligible groups, the baseline branch contains 741,697 available parameter tests and the imputed branch contains 750,603. Observation-row counts and parameter-test counts are therefore different quantities.

### 3.4.3 CCME-WQI Components

The CCME-WQI uses three main factors.

#### F1 — Scope

F1 represents the proportion of selected parameters that fail the water-quality objectives at least once during the assessment period.

$$F_1 = \frac{N_{\text{failed parameters}}}{N_{\text{available parameters}}} \times 100 \tag{1}$$

#### F2 — Frequency

F2 represents how frequently individual measurements fail the objectives.

$$F_2 = \frac{N_{\text{failed tests}}}{N_{\text{total tests}}} \times 100 \tag{2}$$

#### F3 — Amplitude

F3 represents the magnitude by which failed observations exceed their corresponding objectives.

For each failed observation, excursion is calculated. For an upper-limit objective:

$$\text{Excursion} = \frac{\text{Value}}{\text{Objective}} - 1 \tag{3}$$

***Excursion — lower pH limit***

$$\text{Excursion} = \frac{\text{Lower Limit}}{\text{Value}} - 1 \tag{4}$$

***Excursion — upper pH limit***

$$\text{Excursion} = \frac{\text{Value}}{\text{Upper Limit}} - 1 \tag{5}$$

The normalized sum of excursions is then used to get F3.

$$\text{nse} = \frac{\sum \text{Excursions}}{N_{\text{total tests}}} \tag{6}$$

$$F_3 = \frac{\text{nse}}{0.01\text{nse} + 0.01} \tag{7}$$

The same CCME-WQI formulation and water-quality objectives applied consistently to both the baseline and imputed datasets to ensure that the differences in subsequent modelling results could be attributed to preprocessing strategy rather than some changes in the WQI formulation.

### 3.4.4 CCME-WQI Calculation

The three factors are combined using the CCME-WQI formulation:

$$\text{CCME-WQI} = 100 - \frac{\sqrt{F_1^2 + F_2^2 + F_3^2}}{1.732} \tag{8}$$

The resulting value ranges from 0 to 100, where higher values indicates some better water-quality conditions.

### 3.4.5 WQI Classification

**Table 4** CCME-WQI classification

| CCME-WQI | Category |
|---|---|
| 95–100 | Excellent |
| 80–94 | Good |
| 65–79 | Fair |
| 45–64 | Marginal |
| 0–44 | Poor |

## 3.5 Predictor and Target Definition

The calculated CCME_WQI is used as continuous regression target for the subsequent predictive models.

The input features used for the model development were: Biochemical Oxygen Demand (BOD), Dissolved Oxygen (DO), Ammonia Nitrogen (Ammonia-N), Nitrite Nitrogen (Nitrite-N), Nitrate Nitrogen (Nitrate-N), pH.

### 3.5.1 Machine Learning Dataset Preparation

StationCode, Year, and Category were not used as the predictive input features.

The model-preparation script computes the arithmetic mean of each of the six predictors within each StationCode-Year group in the core dataset. These same core-derived predictor means are inner-joined separately with the baseline and imputed WQI targets on StationCode and Year. The supplied implementation does not recompute ML predictor means from the imputed raw measurements; the two modelling datasets differ through their WQI targets and eligible station-year joins. Missing values are excluded from each parameter mean by the aggregation operation. The DNN and TabNet scripts additionally remove any rows with missing predictor or target values before splitting.

The resulting saved ML datasets contain 12,497 and 12,503 rows, respectively, with no missing values in the six predictors or target and no duplicate StationCode-Year keys.

## 3.6 Machine Learning Models

Random Forest [4] and Extra Trees combine predictions from multiple decision trees, with Extra Trees introducing greater randomness during tree construction.

Gradient Boosting builds the trees sequentially, with each new tree focuses on reducing the errors of the made by previous trees. XGBoost provides an optimized

implementation of gradient boosting for structured data [3]. CatBoost is another boosting-based method included in the comparative evaluation [2].

Each ML model is fitted separately on the baseline and imputed training subsets using the six predictors. The scripts pass the predictor values directly to the estimators, without a scaling stage. Table 5 lists every explicitly supplied constructor argument. Parameters not specified in these scripts are left to the installed library defaults; their numerical values and library versions are not recorded in the supplied implementation and are not inferred here.

**Table 5** Explicitly implemented machine-learning configurations

| Model | Explicit configuration |
|---|---|
| Random Forest | `n_estimators=100`, `random_state=42`, `n_jobs=-1`. |
| Extra Trees | `n_estimators=100`, `random_state=42`, `n_jobs=-1`. |
| Gradient Boosting | `n_estimators=100`, `learning_rate=0.1`, `max_depth=3`, `random_state=42`. |
| XGBoost | `n_estimators=300`, `learning_rate=0.05`, `max_depth=6`, `subsample=0.8`, `colsample_bytree=0.8`, `objective="reg:squarederror"`, `random_state=42`, `n_jobs=-1`. |
| CatBoost (holdout) | `iterations=300`, `learning_rate=0.05`, `depth=6`, `loss_function="RMSE"`, `verbose=0`, `random_seed=42`. |

## 3.7 Deep Learning Models

TabNet was included as an attention-based neural model for tabular data [5].

### 3.7.1 Implemented DNN Architecture and Training

The DNN uses a TensorFlow/Keras Sequential architecture with a six-feature input, four Dense hidden layers containing 128, 64, 32, and 16 units, respectively, each with ReLU activation, and a single Dense output unit without an explicitly specified activation. No dropout, batch-normalization layer, or explicit regularization is included. A StandardScaler is fitted only on the training predictors and is then applied to the test predictors; the target is not scaled.

The model is compiled with `optimizer="adam"`, `loss="mse"`, and `metrics=["mae"]`. Training uses `validation_split=0.20`, `epochs=100`, `batch_size=32`, and `verbose=1`. The validation fraction is taken from the training subset rather than from the holdout test subset. No callbacks, early stopping, explicit optimizer learning rate, TensorFlow/NumPy seed, or explicit training-shuffle argument are specified. Consequently, the fixed split seed does not establish seeded neural-network initialization.

### 3.7.2 Implemented TabNet Architecture and Training

The TabNetRegressor is instantiated with `n_d=16`, `n_a=16`, `n_steps=5`, `gamma=1.5`, `lambda_sparse=1e-4`, `optimizer_fn=torch.optim.Adam`, `seed=42`, and `verbose=10`. Other architecture and optimizer settings, including the optimizer learning rate, are not explicitly configured. A StandardScaler is fitted on the training predictors and applied to the test predictors, after which both predictor arrays are converted to `float32`. Targets are reshaped into single-column arrays and are not scaled.

Training specifies `max_epochs=100`, `patience=15`, `batch_size=256`, `virtual_batch_size=128`, `num_workers=0`, and `drop_last=False`. The evaluation

sets are the training subset and the same 20% test subset, named `train` and `test`, with `eval_metric=["mae"]`. The test subset is thus used for early-stopping evaluation as well as final performance reporting; it is not an independent untouched test set for this implementation. No separate validation split or explicit training-shuffle argument is supplied.

## 3.8 Experimental Setup and Data Splitting

All five ML scripts and both DL scripts call `train_test_split` with `test_size=0.20` and `random_state=42`. No station grouping, chronological separation, or stratification is supplied. The split is therefore a random station-year row split, not a held-out-station or future-year evaluation.

For the baseline dataset we have: Training samples: 9,997; Testing samples: 2,500.

For the imputed dataset we have: Training samples: 10,002; Testing samples: 2,501.

## 3.9 Model Evaluation

**Mean Absolute Error (MAE):** measures the average absolute difference between the predicted and actual WQI values.

**Root Mean Squared Error (RMSE):** measures the prediction error while giving greater weight to larger errors.

**Coefficient of Determination ($R^2$):** indicates how well the model explains the variation in the target variable.

Separate CatBoost fits are used in the holdout comparison, five-fold validation, SHAP analysis, and recommendation modules; the validation fit is not assumed to be the identical fitted artifact used in all subsequent components.

## 3.10 CatBoost Cross-Validation

Cross-validation is performed independently on each complete ML dataset, not only on the 80% holdout-training subset. The splitter is `KFold(n_splits=5, shuffle=True, random_state=42)`. For each fold, a new CatBoostRegressor is fitted. Its configuration is `iterations=300`, `depth=6`, `learning_rate=0.05`, `loss_function="RMSE"`, `random_seed=42`, and `verbose=False`.

Each fold yields MAE, RMSE calculated as the square root of mean squared error, and $R^2$. The script summarizes the five fold values with `np.mean` and sample standard deviation using `np.std(ddof=1)`. It does not group folds by station or order them chronologically. This procedure measures variation across shuffled station-year row partitions; it does not establish generalization to previously unseen stations, external basins, or future years.

## 3.11 Explainable AI Using SHAP

SHapley Additive exPlanations (SHAP) describe feature contributions to the fitted CatBoost predictions of the constructed station-year CCME-WQI [1]. The six explained predictors are BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH. Neither type identifies a physical cause of water-quality change.

The figures in this section are saved outputs of `04_SHAP/01_shap_analysis.py` and `04_SHAP/02_local_shap_analysis.py`. Both scripts exclude records missing any predictor or target, use an 80:20 train–test split with `random_state=42`, fit CatBoost

with 300 iterations, depth 6, learning rate 0.05, RMSE loss, and `random_seed=42`, and compute `shap.TreeExplainer` attributions on the test predictors.

## 3.12 Decision-Support Recommendation Framework

### 3.12.1 Model Prediction and Water-Quality Category

The module processes the baseline and imputed machine-learning tables independently. It requires the six predictors BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH, together with the constructed `CCME_WQI` target. Records with a missing value in any of these seven columns are excluded from this module. A separate CatBoost regressor is fitted to all remaining records in each table, saved as a dataset-specific model, and used to predict those same records. Thus, the recommendation outputs generated by this script are in-sample decision-support outputs, not a new holdout evaluation or a forecast of future water quality.

Its category, stored as `Predicted_Category`, is assigned by testing thresholds in descending order: Excellent for a prediction at least 95; Good for a prediction at least 80 but below 95; Fair for a prediction at least 65 but below 80; Marginal for a prediction at least 45 but below 65; and Poor for a prediction below 45. The category function returns Unknown for a missing prediction. The script does not clip predictions before categorization.

### 3.12.2 Guideline Checks and Parameters Requiring Attention

Table 6 records the numerical conditions implemented in the guideline dictionary. A measurement equal to a boundary is within the implemented guideline. Missing values receive the status Missing, although the complete-case selection above removes records with missing predictors before recommendations are generated.

**Table 6** Implemented parameter-level guideline conditions in the recommendation module.

| Parameter | Exceedance condition | Status assigned |
|---|---|---|
| BOD | Value $>$ 3.0 mg/L | Above guideline |
| Ammonia-N | Value $>$ 0.05 mg/L | Above guideline |
| Nitrite-N | Value $>$ 0.5 mg/L | Above guideline |
| Nitrate-N | Value $>$ 0.5 mg/L | Above guideline |
| pH | Value $<$ 6.2 | Below guideline |
| pH | Value $>$ 7.3 | Above guideline |
| DO | No threshold implemented | Within guideline for a nonmissing value |

For DO, this status is a software fallback because no DO guideline exists in the dictionary, not evidence of compliance with an oxygen standard.

For each record, the script constructs an internal `exceeded_parameters` list by checking the guideline-defined parameters in the order BOD, Ammonia-N, Nitrite-N, Nitrate-N, and pH. A parameter is appended once if its status is Above guideline or Below guideline. The list therefore identifies the parameters requiring attention on the basis of threshold violations, and its length counts distinct exceeded parameters rather than individual measurement tests. DO is not included in this list.

### 3.12.3 Exact Monitoring-Priority Rules

Let $E$ denote the number of parameters in the internal exceedance list. The priority function applies the following rules in order, returning as soon as a condition is satisfied:

1. A predicted category of Poor produces HIGH priority.
2. A predicted category of Marginal produces HIGH priority.
3. Any remaining category with $E \geq 2$ produces HIGH priority.
4. A remaining predicted category of Fair produces MEDIUM priority.
5. All remaining cases produce LOW priority.

The generic fallback also assigns LOW to Unknown when $E < 2$. SHAP magnitudes, the selected advice parameter, and the actual category do not enter this priority function. In particular, a single exceedance does not automatically imply MEDIUM or HIGH priority.

### 3.12.4 SHAP-Based Advice Selection and Pressure Interpretation

The script computes local attributions for all six predictors using `shap.TreeExplainer` on the fitted CatBoost model. For each record, it collects features with strictly negative SHAP values and sorts them by their signed values in ascending order. If this collection is nonempty, the feature with the most negative SHAP value becomes the selected advice parameter, and its attribution is recorded. Equal negative values retain the predictor order BOD, DO, Ammonia-N, Nitrite-N, Nitrate-N, and pH.

If no negative attribution exists but the exceedance list is nonempty, the first parameter in that list is selected and its recorded SHAP value is missing. If neither condition applies, the selected parameter is None and its recorded SHAP value is missing.

Table 7 reproduces the implemented mappings. This is deterministic selection of predefined text, not free-form text generation. A selected negative SHAP contributor need not exceed its guideline and need not be a member of the threshold-based attention list. DO can therefore select oxygen-related advice through SHAP, despite having no implemented guideline threshold or role in the exceedance count.

When the selected parameter has no dictionary entry, the pressure output is "No major pollution pressure identified by the rule-based system" and the recommendation is "Continue routine monitoring and maintain current water-quality surveillance."

The pressure messages are predefined hypotheses for follow-up, not findings of source attribution.

## 3.13 Streamlit Dashboard Implementation

It accepts six predictor values and estimates the constructed station-year CCME-WQI; it does not implement temporal forecasting or demonstrate a particular inference latency. Inputs should be interpreted in the context of the station-year mean predictors used for model development. The dashboard displays a clipped prediction and category, configured guideline checks, local SHAP contributions, and advisory outputs.

**Table 7** Exact parameter-specific pressure and recommendation messages implemented in the module.

| Selected parameter | Possible pressure text | Recommendation text |
|---|---|---|
| BOD | Possible organic pollution or wastewater-related pressure | Investigate possible organic or wastewater pollution sources and increase monitoring. |
| Ammonia-N | Possible sewage, organic-waste, or nutrient pollution pressure | Investigate possible sewage or organic-waste inputs and increase monitoring of nitrogen pollution. |
| Nitrite-N | Possible nitrogen-cycle disturbance or wastewater-related pressure | Investigate possible wastewater or nitrogen-cycle disturbances and monitor nitrogen compounds. |
| Nitrate-N | Possible nutrient runoff, agricultural, or wastewater-related pressure | Investigate possible nutrient runoff or wastewater sources and strengthen nutrient monitoring. |
| pH | Possible chemical or industrial influence | Investigate possible chemical or industrial influences and verify pH measurements. |
| DO | Possible oxygen-depletion or organic-pollution stress | Increase monitoring for oxygen depletion and investigate possible organic-pollution sources. |

**Table 8** STREAMLIT APPLICATION WORKFLOW

| | |
|---|---|
| 1. Field Input Layer | Sliders / number fields for BOD, DO, NH3-N, NO2-N, NO3-N, and pH. |
| 2. Inference Engine | Passes vector to serialized CatBoost model artifact. |
| 3. Target Output | Displays CCME-WQI score [0–100] & quality tier. |
| 4. Guideline Checker | Compares values against study-defined water-quality objectives. |
| 5. SHAP Visualizer | Displays signed local SHAP contributions as a bar chart and table. |
| 6. Triage Advisory | Displays dashboard monitoring priority and follow-up advice. |

1. Local Model Explanation: The uploaded dashboard script does not generate waterfall or force plots; the manuscript waterfall figures come from the separate local-analysis script.
2. Advisory Outputs: The dashboard generates advisory outputs based on two primary conditions: the predicted category and whether study-defined objectives have been exceeded. In parallel, the offline Recommendation Engine applies local SHAP attributions to identify a parameter-specific advisory message whenever a negative attribution is present. Although both components are guided by the same overarching monitoring-priority framework, the procedures used to determine the final advisory output are implemented independently.

# 4 Results

## 4.1 Dataset and Missing-Value Results

The processed dataset has the following missing observations:

Following imputation, a validation step was performed to confirm that the process had not produced any negative values. The resulting parameter ranges and summary statistics were subsequently examined in relation to the available measurements to evaluate the characteristics of the completed dataset prior to WQI construction. Importantly, this validation was intended to assess the behavior and plausibility of

**Table 9** Missing observations in the processed dataset

| Parameter | Missing observations | Missing (%) |
|---|---|---|
| BOD | 24,351 | 13.87% |
| DO | 24,578 | 14.00% |
| Ammonia-N | 948 | 0.54% |
| Nitrite-N | 11,153 | 6.35% |
| Nitrate-N | 11,743 | 6.69% |
| pH | 20,314 | 11.57% |

the imputed observations rather than to demonstrate formal equivalence between the imputed and observed distributions.

## 4.2 WQI Results

Table 10 distinguishes raw rows, builder-input rows, eligible-group observation rows, and final station-year records. Reapplying the eligibility conditions to the supplied inputs reproduces the saved output keys and the summed Observations columns exactly.

**Table 10** Observation and station-year counts through dataset construction

| Stage / Count unit | Baseline | Imputed |
|---|---|---|
| Original raw observation rows | 178,680 | 178,680 |
| Rows excluded by all-six-missing preprocessing filter | 3,065 | 0 |
| Observation rows supplied to WQI builder | 175,615 | 178,680 |
| Station-year groups before eligibility checks | 15,502 | 16,639 |
| Groups excluded for fewer than ten input rows | 2,327 | 3,452 |
| Groups excluded for insufficient WQI parameters after passing the row-count check | 678 | 684 |
| Observation rows within eligible groups | 157,701 | 158,965 |
| Final station-year WQI records | 12,497 | 12,503 |

For the baseline branch, 2,327 groups containing 9,150 rows fail the ten-row requirement. Among the remaining groups, 678 groups containing 8,764 rows fail the parameter-availability requirement. The 12,497 eligible groups therefore contain 157,701 observation rows. For the imputed branch, 3,452 groups containing 10,867 rows fail the ten-row requirement; another 684 groups containing 8,848 rows fail the parameter-availability requirement after passing the first check. Its 12,503 eligible groups contain 158,965 observation rows. These excluded groups are skipped when constructing the WQI output; their rows are not removed from the saved core or imputed-input files.

The saved imputed WQI keys contain all 12,497 baseline keys and six additional keys. Each additional group contains nine baseline core rows but ten rows in the original and imputed inputs, and all five WQI parameters are already available before imputation. The six keys are UKGBNIF10325–2003, UKGBNIF10326–2003, UKGBNIF10361–2020, UKGBNIF10363–2020, UKGBNIF10907–2007, and UKGBNIF11451–2020. The six-record increase is therefore explained by retention of an additional all-six-missing row in each group, allowing it to meet the ten-row criterion; it is not evidence that imputation made an additional WQI parameter available in these groups. Applying the same eligibility rules to the original, unimputed raw observations also yields 12,503 eligible groups.

For baseline dataset, the WQI values ranged from 7.07 - 100, with mean of 49.43 and median of 46.66. For imputed dataset, the WQI values ranged from 7.52 - 100, with mean of 49.36 and median of 46.58.

The resulting distribution in the current dataset is:

**Table 11** WQI category distributions

| Category | Baseline | Imputed |
|---|---|---|
| Poor | 5,691 | 5,723 |
| Marginal | 5,044 | 5,021 |
| Fair | 1,189 | 1,190 |
| Good | 554 | 550 |
| Excellent | **19** | 19 |
| Total | **12,497** | 12,503 |

**Table 12** Baseline feature correlations with CCME-WQI

| Feature | Correlation with CCME-WQI |
|---|---|
| Nitrate-N | -0.6997 |
| Nitrite-N | -0.6682 |
| Ammonia-N | -0.3407 |
| pH | -0.2953 |
| BOD | -0.2930 |
| DO | 0.2895 |

The WQI category distributions of baseline and imputed datasets remained broadly similar. The similarity between two distributions indicates the imputation process did not substantially alter the overall WQI classification pattern.

### 4.2.1 Exploratory Analysis for Machine Learning

For the baseline dataset, the mean CCME-WQI was 49.43, with standard deviation of 15.02. For the imputed dataset, the mean CCME-WQI was 49.36, with standard deviation of 15.02.

The imputed dataset showed that nearly identical relationships between the predictor variables and the target variable.

## 4.3 Machine Learning and Deep Learning Results

The benchmark table reports prediction errors and coefficients of determination, but no statistical significance test is documented. Small branch-level metric differences do not by themselves establish preservation of every feature distribution.

**Table 13** Comparative Benchmark Evaluation

| Model Family | Algorithm | Baseline MAE | Baseline RMSE | Baseline $R^2$ | Imputed MAE | Imputed RMSE | Imputed $R^2$ |
|---|---|---|---|---|---|---|---|
| Tree Ensembles | Random Forest | 3.0245 | 4.1346 | 0.9231 | 3.0870 | 4.2095 | 0.9204 |
| | Extra Trees | 3.0872 | 4.1626 | 0.9221 | 3.1329 | 4.1942 | 0.9210 |
| Boosting | Gradient Boosting (GBDT) | 3.0791 | 4.1093 | 0.9240 | 3.1068 | 4.1384 | 0.9231 |
| | XGBoost | 2.9187 | 4.0063 | 0.9278 | 2.9579 | 4.0368 | 0.9268 |
| | CatBoost (Selected) | 2.9632 | 3.9849 | 0.9286 | 2.9799 | 4.0024 | 0.9280 |
| Deep Learning | Deep Neural Net (DNN) | 3.3088 | 4.4594 | 0.9105 | 3.4729 | 4.5312 | 0.9078 |
| | TabNet | 4.8551 | 6.3383 | 0.8193 | 4.6399 | 6.0572 | 0.8352 |

The uploaded project RMSE comparison visualization is reproduced in Figure 2.

Figures 3 and 4 reproduce the baseline and imputed actual-versus-predicted plots generated by `10_prediction_analysis.py`. They visualize prediction agreement on the specified random splits, not geographic transfer or future forecasting.

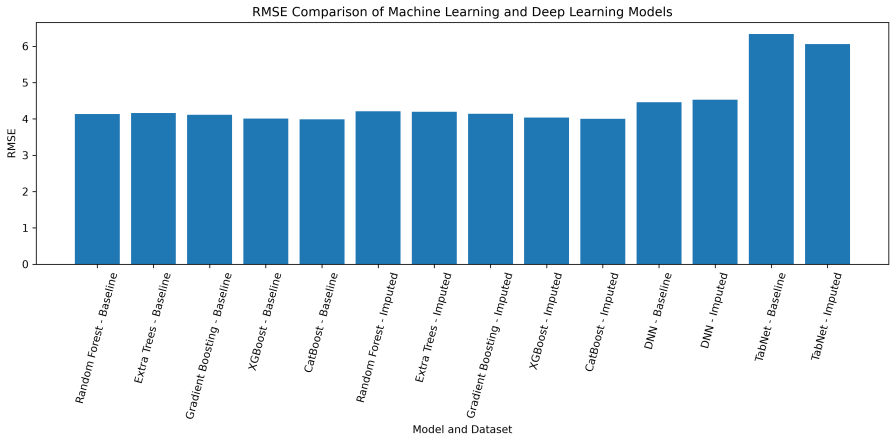



**Fig. 2** Original project RMSE comparison across the five machine-learning models, DNN, and TabNet for baseline and imputed datasets.

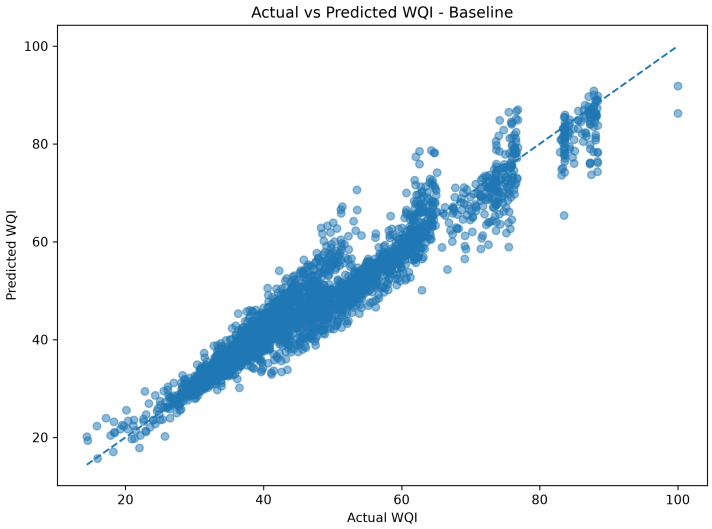



**Fig. 3** Constructed station-year CCME-WQI versus CatBoost prediction for the baseline test set. The dashed line is the identity line; deviations indicate prediction error for individual held-out records.

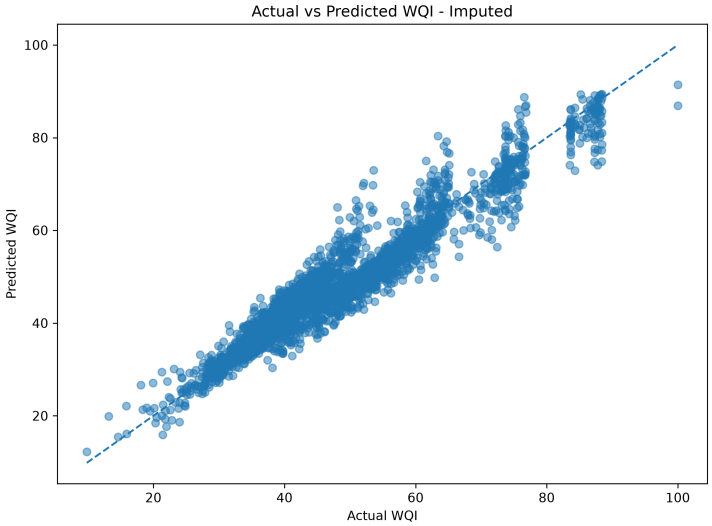



**Fig. 4** Constructed station-year CCME-WQI versus CatBoost prediction for the imputed test set. The dashed line is the identity line. The plotted target is the constructed station-year index, not an independently measured future water-quality outcome.

## 4.4 CatBoost Cross-Validation Results

Baseline Set: Mean MAE = $2.9960 \pm 0.0252$, Mean RMSE = $4.0448 \pm 0.0356$, Mean $R^2 = 0.9274 \pm 0.0018$.

Imputed Set: Mean MAE = $3.0042 \pm 0.0235$, Mean RMSE = $4.0463 \pm 0.0376$, Mean $R^2 = 0.9273 \pm 0.0031$.

## 4.5 SHAP Explainability Results

### 4.5.1 Global Contributions Across Evaluated Samples

Global importance is calculated as the mean absolute SHAP value for each predictor over the evaluated test samples: 2,500 baseline records and 2,501 imputed records, as verified in the saved SHAP-value tables. Figure 5 presents these aggregate magnitudes, while Figure 6 shows the distribution of signed contributions and corresponding predictor values. Mean absolute importance measures attribution magnitude, not its direction or environmental importance.

The summary plots show higher nitrate and ammonia values predominantly associated with negative contributions, and lower values predominantly associated with positive contributions in these evaluated samples. DO has the lowest mean absolute attribution in both tables; this does not establish that oxygen is environmentally unimportant.

The saved mean absolute SHAP tables for the baseline and imputed test samples support the following ranking of model attribution, shown in Figure 5:

$$\text{Nitrate-N} > \text{Ammonia-N} > \text{BOD} > \text{Nitrite-N} > \text{pH} > \text{DO}$$

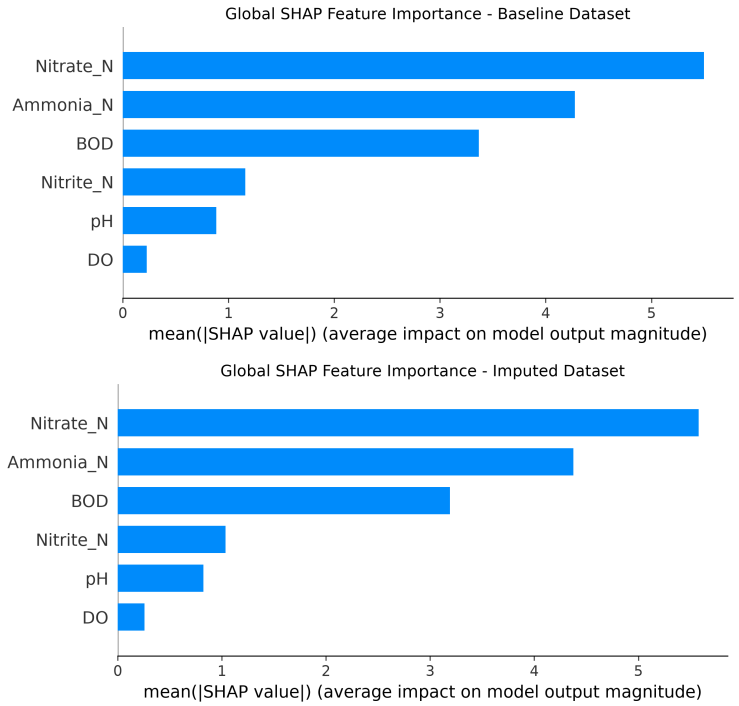



**Fig. 5** Global mean absolute SHAP importance for the baseline (top) and imputed (bottom) CatBoost explanation models, evaluated over 2,500 and 2,501 test records, respectively. Bar lengths summarize contribution magnitudes in predicted WQI units and do not indicate causal effects.

### 4.5.2 Nitrate Dependence in the Fitted Models

Figure 7 provides a complementary view of the highest-ranked predictor by plotting Nitrate-N values against their local SHAP contributions. Coloring denotes the companion predictor identified on each saved plot. The point distribution describes fitted-model behavior within the evaluated samples; it is not a controlled estimate of a nitrate intervention or evidence of a pollution source.

### 4.5.3 Local Sample-Level Decomposition

For an individual station-year predictor vector $x$, local TreeSHAP decomposes the fitted prediction into the explainer expected output and six feature contributions:

$$\hat{y}(x) = \phi_0 + \sum_{i=1}^{M} \phi_i(x).$$

Here $M = 6$, $\phi_0$ is returned by `explainer.expected_value`, and $\phi_i(x)$ is the signed attribution to predictor $i$. The expected model output is not assumed to equal the observed target mean. These signs do not by themselves establish improvement or deterioration of the actual water body. The local-analysis script selects the records with the lowest and highest *predicted* CCME-WQI within each test set. Figures 8 and 9 reproduce all four saved waterfall explanations. Each panel explains only its selected record; the examples are prediction extremes rather than a random or representative

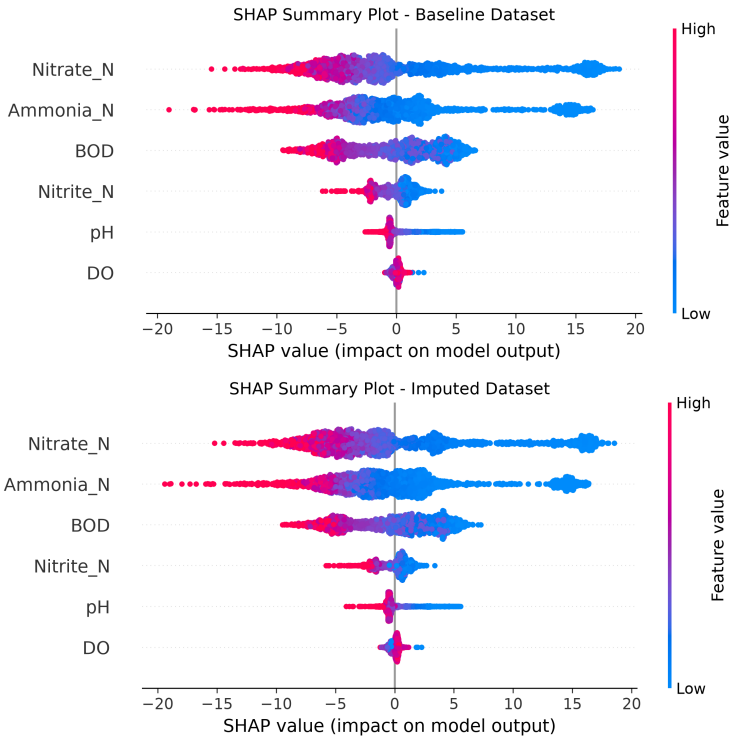



**Fig. 6** Global SHAP summary plots for baseline (top) and imputed (bottom) test records. Each point represents one sample-level attribution; horizontal position indicates the signed contribution relative to the model expected output, and color indicates the relative predictor value. Positive contributions push the prediction upward and negative contributions push it downward.

sample. The saved contribution tables identify Ammonia-N as the largest-magnitude negative contribution in both lowest-prediction examples, and Nitrate-N as the largest positive contribution in both highest-prediction examples. These local observations are distinct from the aggregate ranking.

## 4.6 Recommendation-System Results

For each and every evaluated record, the system provides a predicted WQI, assigns corresponding water-quality category and determines a priority based on predicted category and number of parameters exceeding their specified guideline objectives. SHAP-based local explanations were used to identify the water-quality parameter with the strongest negative influence on each model prediction. This parameter was then linked to predefined advisory messages describing possible contributing pressures and recommended follow-up actions.

## 4.7 Dashboard Validation

To verify the dashboard's functionality under degraded water-quality conditions, a stress-test case was evaluated using the input values: BOD = 8.0 mg/L, DO = 2.0 mg/L, Ammonia-N = 0.5 mg/L, Nitrite-N = 1.0 mg/L, Nitrate-N = 5.0 mg/L, and pH = 8.5. For the selected stress-test case, the dashboard predicted a CCME-WQI

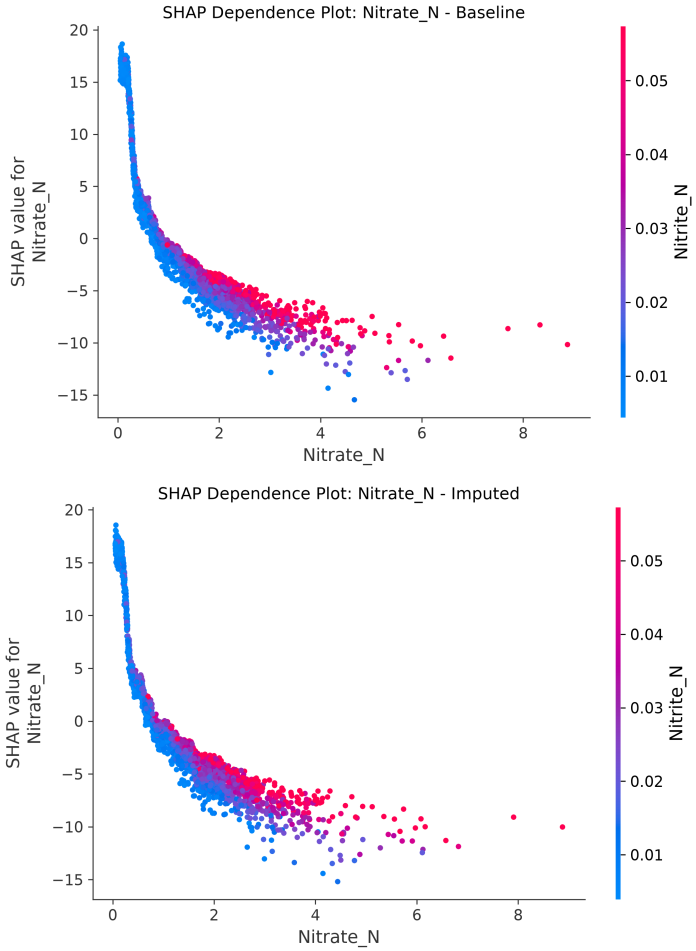



**Fig. 7** Nitrate-N SHAP dependence plots for baseline (top) and imputed (bottom) test samples. The horizontal axis is the observed predictor value and the vertical axis is its contribution to the prediction. Color represents the companion predictor labeled in the original plots and does not establish a causal interaction.

of approximately 19.61, placing the water quality in the Poor category. The system assigned a High monitoring priority, with all five guideline-assessed parameters exceeding their configured thresholds. It also successfully generated the local SHAP explanation along with the associated pressure and recommendation messages. However, this test represents a functional validation of the implemented workflow and should not be considered field validation or evidence of real-world environmental performance.

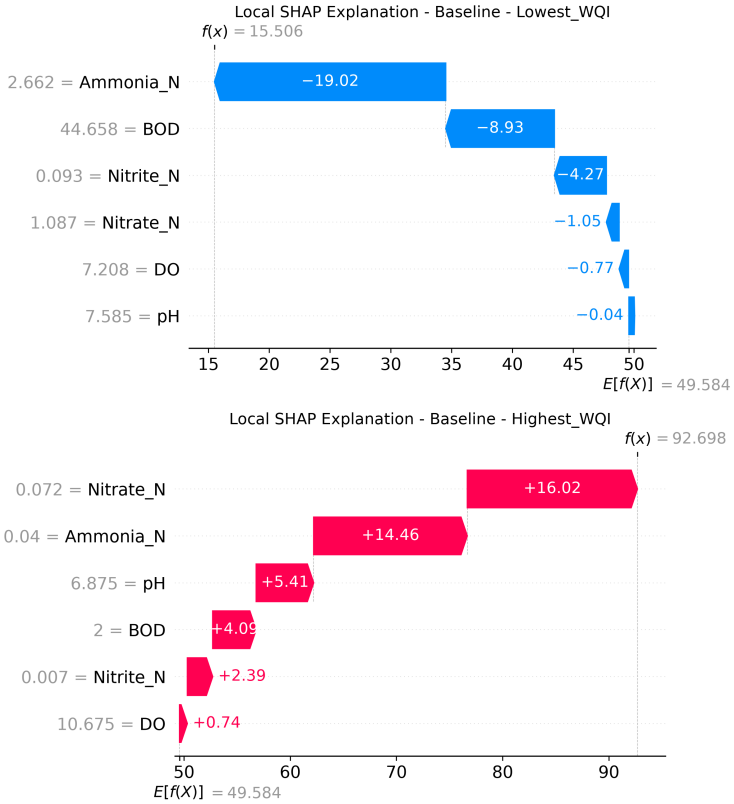



**Fig. 8** Local SHAP waterfall explanations for the baseline test records with the lowest (top) and highest (bottom) predicted WQI. The six signed contributions bridge the expected model output $E[f(X)]$ and individual prediction $f(x)$.

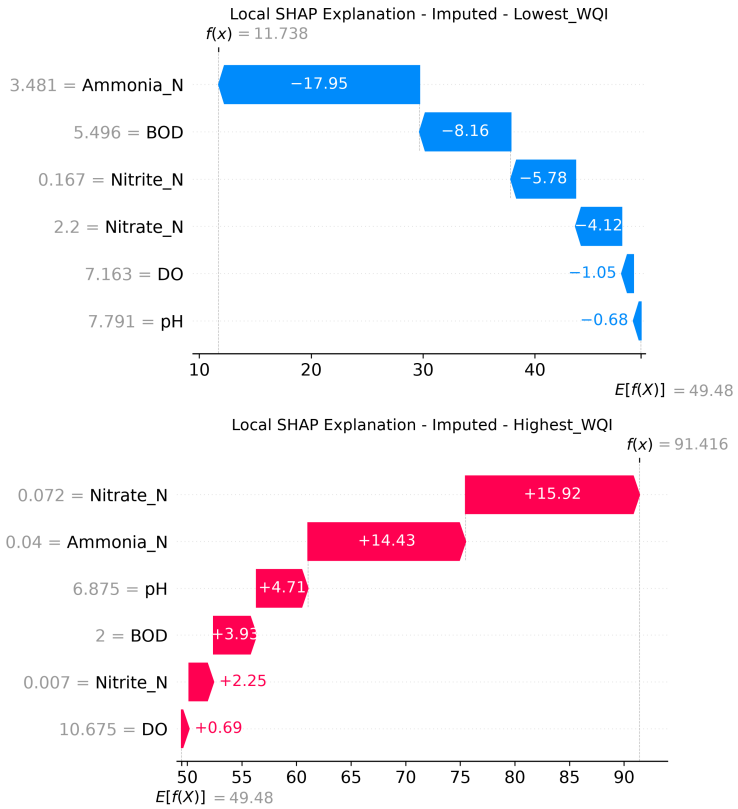



**Fig. 9** Local SHAP waterfall explanations for the imputed test records with the lowest (top) and highest (bottom) predicted WQI. Each panel describes a single prediction; attribution magnitudes do not identify physical causes or verified pollution sources.

# 5 Discussion

## 5.1 Justification for Selecting CatBoost

Although XGBoost achieved a marginally lower Mean Absolute Error (2.9187 vs. 2.9632), the manuscript benchmark reports CatBoost with the lowest overall Root Mean Squared Error (3.9849 baseline, 4.0024 imputed) and the highest coefficient of determination ($R^2 = 0.9286$). RMSE weights larger prediction errors more heavily; it does not establish detection of hazardous events. CatBoost was selected for later analysis. No comparative SHAP-timing experiment or isolated feature-interaction evaluation is supplied, so faster explanation computation and superior interaction handling are not claimed.

## 5.2 System Constraints and Future Extensions

1. Parameter Scope: The model operates strictly over 6 chemical parameters. In real river basins, variables such as water temperature, electrical conductivity, seasonal discharge rate, heavy metals, and upstream land use provide valuable supplementary signals.
2. Data-Quality Scope: The project processes historical monitoring observations. The experiments do not establish performance under live sensor drift or transmission loss, and no sensor-calibration evaluation is reported.
3. Target Construction and Predictor Relationship: The CCME-WQI used as the prediction target is a constructed composite index based on five of the water-quality parameters included in the predictor set. Therefore, the reported model performance should be interpreted as estimation of the constructed WQI from its underlying measurements rather than prediction of an independently measured environmental outcome. The high predictive performance should consequently not be interpreted as evidence of equivalent performance for independently observed or future water-quality outcomes.
4. Advisory Scope: The recommendation engine generates hypothesis-driven advisories based on parameter patterns. These guidelines assist human inspectors in prioritizing field visits but do not replace certified laboratory testing.

### 5.2.1 Future Roadmap

Geospatial Plume Tracking: Incorporating GIS coordinates into the Streamlit dashboard to visualize downstream pollutant diffusion.

Temporal Forecasting: Future work could investigate sequence models with explicitly defined future targets and chronological validation. The current implementation has neither a forecasting task nor a validated forecast horizon.

External Cross-Basin Validation: Testing model portability against river monitoring datasets from other geographic regions.

# 6 Conclusion

This study develops an explainable AI-based decision-support system for estimating and interpreting the constructed station-year CCME-WQI from six predictors. It does not forecast future water quality. It uses long-term Northern Ireland Environment

Agency (NIEA) observations and includes preprocessing, missingness analysis, WQI construction, model comparison, row-based validation, SHAP analysis, recommendations, and dashboard development.

The ML model Cat-Boost showed strong overall predictive performance and was selected for the later stages of the system. Five-fold cross-validation is helpful to show that the model produced consistent results across different validation folds.

SHAP was then used to explain the predictions by observing the contribution of the input parameters. By combining the predicted WQI, guideline exceedance information, and model interpretation we have developed A recommendation engine.

The Final system is implemented as an interactive Streamlit dashboard.

The main value of the proposed system is the integration of prediction, explanation, and decision support into a single workflow, Rather than providing only a numerical WQI prediction.

# Declarations

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**Competing Interests.** The authors declare no competing interests.

**Ethics Approval and Consent to Participate.** Not applicable. This study used an environmental water-quality monitoring dataset and did not involve human participants, human subjects, or animal experimentation.

**Consent for Publication.** Not applicable.

**Data Availability.** Availability of the underlying dataset is subject to the terms and conditions under which the source data were obtained.

**Materials Availability.** Not applicable.

**Code Availability.** A public repository is not specified at the time of manuscript preparation.

**Author Contributions.** Chintakayala Karthikeya: Conceptualization, methodology, data analysis, model development, experimentation, and manuscript preparation.

Ameer Roshan Shaik: Data preprocessing, implementation, experimentation, and manuscript review.

Shaik Abdul Samad: Machine-learning/deep-learning implementation, explainable AI analysis, dashboard development, and manuscript preparation.

Desam Pranay Kumar Reddy: Data analysis, recommendation-system development, visualization, experimentation, and manuscript review.

All authors read and approved the final manuscript.

# References

[1] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, pp. 4765–4774, 2017.

[2] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: Unbiased Boosting with Categorical Features," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, pp. 6638–6648, 2018.

[3] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794, 2016.

[4] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[5] S. O. Arik and T. Pfister, "TabNet: Attentive Interpretable Tabular Learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 8, pp. 6679–6687, 2021.

[6] W. Li, Y. Zhao, Y. Zhu, Z. Dong, F. Wang, and F. Huang, "Research progress in water quality prediction based on deep learning technology: a review," *Environmental Science and Pollution Research*, vol. 31, pp. 26415–26431, 2024.

[7] X. Yan, T. Zhang, W. Du, Q. Meng, X. Xu, and X. Zhao, "A Comprehensive Review of Machine Learning for Water Quality Prediction over the Past Five Years," *Journal of Marine Science and Engineering*, vol. 12, no. 1, Article 159, 2024.

[8] R. Choudhary et al., "Predicting water quality index using stacked ensemble regression and SHAP based explainable artificial intelligence," *Scientific Reports*, vol. 15, Article 31139, 2025.

[9] K. Pandey et al., "A systematic review on machine learning deep learning and IoT for water quality prediction," *Discover Sustainability*, vol. 7, Article 1279, 2026.

[10] J. Malisaba, E. B. Omer, E. Bubu, et al., "A review of advanced machine learning and deep learning technologies for water quality management," *Discover Applied Sciences*, vol. 8, Article 670, 2026.

[11] OpenDataNI. *River Water Quality Monitoring 1990 to 2024 – All Parameters*. Dataset catalogue describing Northern Ireland Environment Agency water-quality database records. https://www.data.gov.uk/dataset/0840ab33-18ae-4670-8523-ac43105f5902/river-water-quality-monitoring-1990-to-2024-all-parameters. Accessed 30 September 2026.

[12] Canadian Council of Ministers of the Environment. *Canadian Water Quality Guidelines for the Protection of Aquatic Life: CCME Water Quality Index User's Manual, 2017 Update*. Canadian Environmental Quality Guidelines, 2017. https://ccme.ca/en/res/wqimanualen.pdf.

[13] Project implementation files supplied for manuscript verification: `05_build_station_year_WQI.py` in `02_Models/04_WQI_Calculation/`, and `05_build_station_year_WQI_imputed.py` in `02_Models/04_WQI_New/`. Unpublished Python scripts documenting the configured objectives and eligibility rules; release version and public archival identifier not supplied.